LUIS CARVALHO

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Cloud and Data Engineer with 7+ years of engineering experience, specializing in AWS, Python, SQL, IoT data pipelines and real-time data processing. Experienced in designing and maintaining data ingestion and ETL workflows that transform, validate, enrich and store telemetry from thousands of connected devices.

Hands-on experience with AWS IoT Core, AWS Lambda, Amazon EC2, Amazon S3, Amazon RDS, AWS Aurora, MySQL, MQTT and Kafka. Strong background in data modelling, database architecture, query optimization, data retention, and the integration of edge computing systems with cloud platforms.

Combines cloud infrastructure, backend development and data engineering experience to deliver reliable data solutions, APIs and dashboards for industrial and smart-city applications. Comfortable working across the data lifecycle, from ingestion and processing to storage and data access.

TECHNICAL SKILLS

- **Cloud & Hosting:** AWS (AWS IoT Core, AWS Lambda, Amazon EC2, Amazon S3), Huawei cloud services
- **Programming & Scripting:** Python, PHP, JavaScript, HTML, CSS, SQL, C, Bash/Shell
- **Data Engineering & ETL:**
  - Design and implementation of data pipelines
  - Data ingestion, transformation, validation and enrichment
  - ETL and event-driven data processing
  - Real-time IoT telemetry processing
  - Structured and semi-structured data handling
  - Data integration and storage workflows
- **Development Methodologies:** Agile and Waterfall methodologies
- **Databases & Data Architecture:** MySQL, Amazon RDS and AWS Aurora IoT, Data modelling, schemas and relational database design, Table relationships and data normalization, Indexing and query optimization
- **Infrastructure & Development:** Linux, Docker, Raspberry Pi, NVIDIA ecosystem, Postman, Web application development and deployment
- **Streaming & Integration:** REST APIs, MQTT, Apache Kafka, Event-driven processing, Device-to-cloud integration
- **Soft Skills:** Analytical Thinking, Problem-Solving, Adaptability, Communication, Technical Ownership

PROFESSIONAL EXPERIENCE

2019–PRESENT

Cloud & Infrastructure Engineer / Omniflow S.A.

Progressed from R&D engineering into cloud and infrastructure responsibilities in 2023, working on IoT platforms and data-driven solutions for smart-city and industrial applications.

- Designed and maintained real-time IoT data pipelines for ingesting, processing and storing telemetry from thousands of connected devices, using MQTT, AWS IoT Core, AWS Lambda and SQL databases.
- Developed AWS Lambda functions for event-driven IoT data processing, implementing data normalization, validation and enrichment before database persistence to improve data consistency and support downstream applications.
- Worked with AWS services such as AWS IoT Core, AWS Lambda, Amazon EC2, and Amazon S3 to help with IoT data ingestion, processing, and storage. Worked with MQTT-based device communication and Kafka pipelines connecting NVIDIA-based systems, supporting the movement of data between edge computing environments and downstream processing systems.
- Designed data models and storage architecture for lighting, environmental monitoring, video analytics and energy management workloads. Worked with schemas, table relationships, normalization, indexing, partitioning, data retention and storage selection.
- Developed and maintained MySQL-based data systems, working with Amazon RDS and AWS Aurora alongside other database technologies. Applied database design and query optimization practices to support data access and application requirements.
- Developed and integrated REST APIs to make device information and processed data available to applications, dashboards and other systems.
- Worked with Linux, Docker, Raspberry Pi and NVIDIA Jetson environments to support data collection, edge processing and cloud integration. Data products and visualization: Designed and developed web-based platforms and dashboards, including Omniflow SCADA and the Omniflow Dashboard, enabling users to explore real-time and historical data from connected infrastructure.
- Contributed to R&D and customer-oriented projects combining hardware, software, cloud infrastructure, data processing and visualization.

2025–PRESENT

WEB DEVELOPMENT / Freelance Work

- **Website and E-commerce Development:** Developed a website for a plastic products manufacturer to showcase its product range and provide basic e-commerce functionality.
- **IoT Water Level Monitoring:** Designed and built a water-level monitoring system using an ESP32. Implemented email alerts when the water level dropped below a threshold, allowing the user to activate a water pump when needed.
- **Custom Inventory Management System:** Developed a custom inventory management application using PHP and MySQL, including product registration, stock tracking, and stock movement management.
- **End-to-End Website Development:** Managed the complete website development process, from initial requirements and implementation to deployment and maintenance. Worked with PHP, JavaScript, HTML, CSS, WordPress, and Wix, depending on project requirements.

TECHNICAL PROJECT

Edge AI, IoT Data Processing & Cloud Integration

Worked with NVIDIA Jetson hardware to deploy object-detection models across multiple cameras, collect metrics, and monitor air quality. Integrated the edge devices with AWS IoT Core to transmit collected data to the cloud.

Store incoming data in a database and used AWS Lambda to process it and execute functions in response to incoming data. Also used Amazon EC2 with the NVIDIA TAO Toolkit to train object-detection models.

Project reference: https://www.omniflow.io/post/smart-lampposts-nvidia-aws

IoT Dashboard for smart city management Omniflow

Contributed to the development of a dashboard for monitoring smart city infrastructure through real-time and historical data. The application includes an interactive 3D map that provides a visual overview of infrastructure and allows users to explore different layers of information.

The dashboard displays data such as traffic patterns, air quality, and energy consumption, helping users identify trends and better understand the performance of urban infrastructure.

Dashboard: omniflow.online/Dashboard

EDUCATION

2023-IN PROGRESS
Master's Degree in Critical Computer Systems / Instituto Superior de Engenharia do Porto
2014–2021
Bachelor's Degree in Electrical and Computer Engineering / Instituto Superior de Engenharia do Porto

LANGUAGES & ADDITIONAL INFORMATION

Portuguese: Native | **English:** Intermediate | **Spanish:** Basic
Driving license: Category B